

UEI607 : DIGITAL SIGNAL PROCESSING AND APPLICATIONS

L	T	P	Cr
3	1	2	4.5

Course Objective: To understand the basic concepts and techniques of digital signal processing, familiarization with DSP concepts by studying the design of different digital filters and transform- domain processing.

Introduction: Review of Discrete Time Signals and Systems and z-Transforms, Solution of Difference Equations using One-sided z-Transform, Frequency domain Characteristics of LTI Systems, LTI Systems as Frequency-Selective Filters.

Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT): Discrete-time Fourier Transform, Discrete Fourier Transform and its Properties, Divide and Conquer Approach, Decimation in Time and Decimation in Frequency FFT Algorithms.

Digital Filter Structure: Describing Equation of digital filter, Structures for FIR Systems: Direct Form Structure, Cascade Form Structure, Structure for IIR Systems: Direct Form Structures, Cascade Form Structure, Parallel Form Structure and Lattice Structure.

Design of Digital Filters: Causality and its Implications, Difference between analog filters and digital filters, FIR filter design using windows, Design of IIR filters from analog filters using: Approximation of Derivatives, Impulse Invariance and Bilinear Transformation, Frequency transformations.

Analysis of Finite Word length Effects: Introduction, The quantization process and errors, Input quantization error, product quantization error, Analysis of coefficient quantization effects in FIR filters, A/D noise analysis, Analysis of arithmetic round off errors, Limit cycles in IIR filters.

Laboratory work: Convolution and correlation, Solution of difference equations using z-Transform and Fourier tools, FFT and spectrum analysis, design of high pass, low pass, band pass and band stop FIR filter using window method, design of IIR filter using Matched Z Transform (MZT), Bilinear Z Transform (BZT), Pole Zero Placement and Impulse Invariant methods.

Course Learning Outcomes (CLO): After the completion of the course student will be able to:

1. Analyze signals in both the time and frequency domains to extract relevant information.
2. Apply transformation tools, such as Fourier and Z-transforms, to signals and systems
3. Design and implement structures for different types of digital.
4. Design and analyze various digital filters, focusing on their frequency responses to optimize performance
5. Evaluate the impact of finite word-length effects on digital signal processing

Text Books

1. *Proakis, J.G. and Manolakis, D.G., Digital Signal Processing, Prentice Hall of India Private Limited (2006).*
2. *Rabiner, C.R. and Gold, B., Theory and Applications of Digital Signal Processing, Prentice Hall of India Private Limited (2000).*

Reference Books:

1. *Antonion, A., Digital Filters: Analysis Design and Application, Prentice Hall of India Private Limited (1999).*
2. *Oppenheim, A.V. and Schafer, R.W., Digital Signal Processing, Prentice Hall of India Private Limited (1998).*

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional	30